

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	12 July 2026
Team ID	Buyya Vaishnavi
Project Name	ShopEZ
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Customer Registration through Form Seller Registration through Form
FR-2	User Authentication	Login using Email and Password JWT-based Authentication
FR-3	Product Management	Add Product Details Update Stock and Price Delete Product Listing
FR-4	Product Discovery	Search Products by Category View Product Details
FR-5	Cart & Checkout	Add to Cart Update / Remove Cart Items Place Order
FR-6	Order Management	Update Order Status Cancel Order
FR-7	Dashboard Management	Customer Dashboard Seller Dashboard Admin Dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple and user-friendly interface for customers, sellers, and administrators.
NFR-2	Security	User data must be protected using JWT authentication and secure password hashing (bcryptjs).
NFR-3	Reliability	The system should consistently manage products, carts and orders without failures.
NFR-4	Performance	Search, login, and checkout operations should respond quickly with minimal delay.
NFR-5	Availability	The platform should be accessible whenever users need to browse or purchase products.
NFR-6	Scalability	The system should support increasing numbers of customers, sellers, products, and orders without performance degradation.